Department of Economics
Columbia University

Economics UN3412
Fall 2026, Section 2

Midterm Review Session

Monday October 26, 2026

Practice questions in the format of Wednesday's midterm. Each Stata output is followed by the same regression in R. The long question's outputs are collected in a figure at the end.

1. Short Questions

(1) Columbia's Center for Career Education wants to know whether its one-on-one career coaching raises students' starting salaries. Let Y_i be student i's starting salary and $D_i = 1$ if the student is coached ($= 0$ otherwise). Y_{1i} and Y_{0i} are the student's potential starting salaries with and without coaching. Two analysts study the program.

The first analyst uses a year in which 1,000 seniors applied for 500 coaching slots, and the slots were assigned by lottery. Every lottery winner was coached and no loser was. Winners earn \$3,000 more than losers: $\mathsf{E}[Y_i|D_i = 1] - \mathsf{E}[Y_i|D_i = 0] = \$3{,}000$.

The second analyst compares all coached students with students who never applied, and finds $\mathsf{E}[Y_i|D_i = 1] - \mathsf{E}[Y_i|D_i = 0] = \$9{,}000$.

(a) Which analyst gives a compelling estimate of the causal effect of coaching? Explain mathematically and in words why the other one does not, using $\mathsf{E}[Y_{0i}|D_i = 1]$ and $\mathsf{E}[Y_{0i}|D_i = 0]$.

(b) Suppose coaching raises every student's salary by the same amount, and the first analyst's estimate is exactly right. How large is the selection bias in the second analyst's comparison? Give one reason why it is positive.

(2) A dean wants to know whether joining a study group raises first-year GPA. Let $D_i = 1$ if student i joins a study group ($= 0$ otherwise), and consider the regression model

$$\text{GPA}_i = \beta_0 + \beta_1 D_i + u_i.$$

Assume that joining a study group changes every student's GPA by the same amount, and let Y_{1i} and Y_{0i} be student i's GPA with and without a study group.

(a) Write β_0, β_1 and u_i in terms of the potential outcomes.

(b) What assumption about u_i makes the OLS estimator $\hat{\beta}_1$ unbiased for β_1? State the same assumption in the language of potential outcomes, and explain in words why the two statements say the same thing.

(c) In a very large sample, $\hat{\beta}_1 = 0.30$. Suppose that $\mathsf{E}[u_i|D_i = 1] = 0.15$ and $\mathsf{E}[u_i|D_i = 0] = -0.05$. What is β_1? Does the regression overstate or understate the effect of a study group?

(3) Do graduates of private colleges earn more? Using the College Scorecard data on 988 colleges from class, we regress `earnings`, the median earnings of a college's graduates ten years after they enter, in dollars, on `public`, which equals 1 for public colleges and 0 for private nonprofit colleges.

```
. regress earnings public, robust
Linear regression                               Number of obs     =        988
                                                F(1, 986)         =      53.52
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0447
                                                Root MSE          =      15282
------------------------------------------------------------------------------
             |               Robust
    earnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      public |  -6713.993   917.7193    -7.32   0.000    -8514.901   -4913.086
       _cons |   62372.71   725.6657    85.95   0.000     60948.68    63796.73
------------------------------------------------------------------------------
```

```
model_a <- lm(earnings ~ public, data = scorecard)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
t test of coefficients:
            Estimate Std. Error t value  Pr(>|t|)
(Intercept) 62372.71     725.67  85.952 < 2.2e-16 ***
public      -6713.99     917.72  -7.316 5.289e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Then we add `sat`, the average SAT score of the college's admitted students, and we also regress `sat` on `public`:

```
. regress earnings public sat, robust
Linear regression                               Number of obs     =        988
                                                F(2, 985)         =     334.38
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5072
                                                Root MSE          =      10982
------------------------------------------------------------------------------
             |               Robust
    earnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      public |  -693.1831   630.4174    -1.10   0.272    -1930.299    543.9324
         sat |   74.99688   2.920516    25.68   0.000     69.26573    80.72802
       _cons |  -29142.13   3387.678    -8.60   0.000    -35790.03   -22494.24
------------------------------------------------------------------------------
```

```
model_b <- lm(earnings ~ public + sat, data = scorecard)
coeftest(model_b, vcov = vcovHC(model_b, type = "HC1"))
t test of coefficients:
               Estimate  Std. Error t value Pr(>|t|)
(Intercept) -29142.1337   3387.6780 -8.6024   <2e-16 ***
public        -693.1831    630.4174 -1.0996   0.2718
sat             74.9969      2.9205 25.6793   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
. regress sat public, robust
Linear regression                               Number of obs     =        988
                                                F(1, 986)         =      83.79
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0721
                                                Root MSE          =     141.78
------------------------------------------------------------------------------
             |               Robust
         sat | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      public |  -80.28081   8.770221    -9.15   0.000    -97.49125   -63.07037
       _cons |   1220.249   6.418097   190.13   0.000     1207.654    1232.843
------------------------------------------------------------------------------
```

```
model_c <- lm(sat ~ public, data = scorecard)
coeftest(model_c, vcov = vcovHC(model_c, type = "HC1"))
t test of coefficients:
             Estimate Std. Error  t value  Pr(>|t|)
(Intercept) 1220.2487     6.4181 190.1263 < 2.2e-16 ***
public       -80.2808     8.7702  -9.1538 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) Interpret the coefficient on `public` in the first regression.

(b) Did the first regression suffer from omitted variable bias? Is the correlation between `public` and `sat` positive or negative?

(c) Use the output to calculate the omitted variable bias in the coefficient on `public` in the first regression. Show that it accounts for the difference between the coefficients on `public` in the first two regressions.

(d) Interpret the coefficient on `public` in the second regression.

(4) A university draws 300 of its 600 first-year students at random and pairs each of them with an upper-year mentor. At the end of the year we observe each student's `gpa`, whether the student had a mentor (`mentor` $= 1$) and the student's `sat` score from before college. Consider

$$\text{gpa}_i = \beta_0 + \beta_1 \, \text{mentor}_i + u_i.$$

```
. regress sat mentor, robust
Linear regression                               Number of obs     =        600
                                                F(1, 598)         =       0.15
                                                Prob > F          =     0.7022
                                                R-squared         =     0.0002
                                                Root MSE          =     64.039
------------------------------------------------------------------------------
             |               Robust
         sat | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      mentor |         -2   5.228769    -0.38   0.702    -12.26898    8.268982
       _cons |       1480   3.775176   392.03   0.000     1472.586    1487.414
------------------------------------------------------------------------------
```

```
model_a <- lm(sat ~ mentor, data = mentoring)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
t test of coefficients:
             Estimate Std. Error  t value Pr(>|t|)
(Intercept) 1480.0000     3.7752 392.0347   <2e-16 ***
mentor        -2.0000     5.2288  -0.3825   0.7022
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
. regress mentor sat, robust
Linear regression                               Number of obs     =        600
                                                F(1, 598)         =       0.15
                                                Prob > F          =     0.7022
                                                R-squared         =     0.0002
                                                Root MSE          =     .50077
------------------------------------------------------------------------------
             |               Robust
      mentor | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sat |  -.0001223   .0003197    -0.38   0.702    -.0007501    .0005055
       _cons |   .6808806   .4732376     1.44   0.151    -.2485291     1.61029
------------------------------------------------------------------------------

model_r <- lm(mentor ~ sat, data = mentoring)
coeftest(model_r, vcov = vcovHC(model_r, type = "HC1"))
t test of coefficients:
              Estimate  Std. Error t value Pr(>|t|)
(Intercept)  0.68088055  0.47323755  1.4388   0.1507
sat         -0.00012230  0.00031965 -0.3826   0.7022

. regress gpa mentor, robust
Linear regression                               Number of obs     =        600
                                                F(1, 598)         =      15.51
                                                Prob > F          =     0.0001
                                                R-squared         =     0.0253
                                                Root MSE          =     .32864
------------------------------------------------------------------------------
             |               Robust
         gpa | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      mentor |   .1056667   .0268332     3.94   0.000     .0529679    .1583655
       _cons |   3.358933   .0189988   176.80   0.000     3.321621    3.396246
------------------------------------------------------------------------------

model_b <- lm(gpa ~ mentor, data = mentoring)
coeftest(model_b, vcov = vcovHC(model_b, type = "HC1"))
t test of coefficients:
            Estimate Std. Error  t value  Pr(>|t|)
(Intercept) 3.358933   0.018999 176.7970 < 2.2e-16 ***
mentor      0.105667   0.026833   3.9379 9.188e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

. regress gpa mentor sat, robust
Linear regression                               Number of obs     =        600
                                                F(2, 597)         =      63.36
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1639
                                                Root MSE          =     .30463
------------------------------------------------------------------------------
             |               Robust
         gpa | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      mentor |   .1095369    .024853     4.41   0.000     .0607269    .1583468
         sat |   .0019351   .0001907    10.15   0.000     .0015606    .0023096
       _cons |   .4949728   .2820397     1.75   0.080    -.0589378    1.048883
------------------------------------------------------------------------------

model_c <- lm(gpa ~ mentor + sat, data = mentoring)
coeftest(model_c, vcov = vcovHC(model_c, type = "HC1"))
t test of coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept) 0.4949729  0.2820397  1.7550   0.07978 .
mentor      0.1095369  0.0248530  4.4074 1.241e-05 ***
sat         0.0019351  0.0001907 10.1475 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) Let Y_{1i} and Y_{0i} be student i's GPA with and without a mentor. What are β_0 and β_1 in terms of potential outcomes? Why does the random draw make $\mathsf{E}[u_i|\text{mentor}_i] = 0$?

(b) What does the regression of `sat` on `mentor` check, and what do you conclude?

(c) A classmate runs the regression the other way round: `mentor` on `sat`. Interpret the coefficient on `sat`. Does a student's SAT score affect whether the student gets a mentor? What does this regression check?

(d) Interpret the coefficient on `mentor` in the regression of `gpa` on `mentor`, and test whether mentoring has any effect at the 5% level. Is it the causal effect of a mentor? Explain.

(e) Adding `sat` barely moves the coefficient on `mentor`, but its standard error falls from 0.0268 to 0.0249. Explain both.

(f) Another university makes mentoring voluntary, and finds that students who sign up have a GPA 0.30 points higher than students who do not. Why might this overstate the effect of a mentor? Name the bias and sign it.

(5) Using the same 988 colleges, we estimate three regressions. `admrate` is the share of applicants a college admits, between 0 and 1; `undergrads` is its number of undergraduates; `lnearnings` = ln(`earnings`) and `lnundergrads` = ln(`undergrads`).

```
. regress lnearnings admrate, robust
Linear regression                               Number of obs     =        988
                                                F(1, 986)         =     180.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1870
                                                Root MSE          =     .21803
------------------------------------------------------------------------------
             |               Robust
  lnearnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     admrate |  -.4535513   .0337938   -13.42   0.000    -.5198674   -.3872352
       _cons |   11.28189   .0261431   431.54   0.000     11.23059     11.3332
------------------------------------------------------------------------------

model_a <- lm(lnearnings ~ admrate, data = scorecard)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
t test of coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept) 11.281894   0.026143 431.545 < 2.2e-16 ***
admrate     -0.453551   0.033794 -13.421 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

. regress earnings lnundergrads, robust
Linear regression                               Number of obs     =        988
                                                F(1, 986)         =      42.58
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0415
                                                Root MSE          =      15308
------------------------------------------------------------------------------
             |               Robust
    earnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
lnundergrads |   2745.686   420.7511     6.53   0.000     1920.016    3571.357
       _cons |   37616.22   3423.803    10.99   0.000     30897.44       44335
------------------------------------------------------------------------------

model_b <- lm(earnings ~ lnundergrads, data = scorecard)
coeftest(model_b, vcov = vcovHC(model_b, type = "HC1"))
t test of coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept)  37616.22    3423.80 10.9867 < 2.2e-16 ***
lnundergrads  2745.69     420.75  6.5257 1.081e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
. regress lnearnings lnundergrads, robust
Linear regression                               Number of obs     =        988
                                                F(1, 986)         =      56.36
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0565
                                                Root MSE          =     .23487
------------------------------------------------------------------------------
             |               Robust
  lnearnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
lnundergrads |   .0495782   .0066043     7.51   0.000     .0366182    .0625382
       _cons |   10.56807   .0540871   195.39   0.000     10.46194    10.67421
------------------------------------------------------------------------------

model_c <- lm(lnearnings ~ lnundergrads, data = scorecard)
coeftest(model_c, vcov = vcovHC(model_c, type = "HC1"))
t test of coefficients:
              Estimate Std. Error t value  Pr(>|t|)
(Intercept)  10.5680746  0.0540871 195.390 < 2.2e-16 ***
lnundergrads  0.0495782  0.0066043   7.507 1.351e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) Interpret the coefficient on `admrate`. By about how much do earnings differ between a college that admits 20% of its applicants and one that admits 60%?

(b) Interpret the coefficient on `lnundergrads` in the second regression and in the third.

(c) A classmate reads the third regression and says: "A college that grows its enrollment by 10% raises its graduates' earnings by about 5%." Find two problems with this statement.

(6) Is the payoff of a more selective college the same at public and private colleges? With the same 988 colleges, we regress `earnings` on `public`, on `sat100`, the average SAT score in hundreds of points (`sat100` = `sat`/100), and on their product, `public_sat` = `public` × `sat100`:

```
. regress earnings public sat100 public_sat, robust
Linear regression                               Number of obs     =        988
                                                F(3, 984)         =     277.56
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5089
                                                Root MSE          =      10968
------------------------------------------------------------------------------
             |               Robust
    earnings | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      public |   11185.54   5955.013     1.88   0.061     -500.442    22871.53
     sat100 |   7800.875   386.1558    20.20   0.000     7043.091    8558.658
  public_sat |  -1020.812   517.7153    -1.97   0.049    -2036.765   -4.859067
       _cons |  -32817.37   4520.622    -7.26   0.000    -41688.53    -23946.2
------------------------------------------------------------------------------

model_a <- lm(earnings ~ public + sat100 + public_sat, data = scorecard)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
t test of coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -32817.37    4520.62 -7.2595 7.882e-13 ***
public       11185.54    5955.01  1.8783   0.06063 .
sat100        7800.87     386.16 20.2014 < 2.2e-16 ***
public_sat   -1020.81     517.72 -1.9718   0.04892 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) Write down the estimated regression line for private colleges and for public colleges.

(b) Interpret the coefficients on `sat100` and on `public_sat`.

(c) Is the payoff of 100 SAT points different at public and at private colleges, at the 5% level?

(d) Interpret the coefficient on `public`. Does it make sense on its own?

(7) Students who do a summer internship earn \$8,000 more in their first job than students who do not. In a large sample we estimate

$$\text{Salary}_i = \beta_0 + \beta_1 \text{Intern}_i + u_i, \qquad \hat{\beta}_1 = 8{,}000.$$

Suppose GPA is left out of this regression. Holding internships fixed, each extra point of GPA raises the first salary by \$5,000. Students who do internships have an average GPA of 3.6; students who do not, 3.3.

(a) Use the omitted variable bias formula to compute the bias in $\hat{\beta}_1$. Hint: when X is a dummy, $\text{cov}(X_i, W_i)/\text{var}(X_i)$ is the difference in the mean of W between the two groups.

(b) What would the coefficient on Intern_i be in a regression that also includes GPA?

(c) A friend concludes: "So the causal effect of an internship is \$6,500." Do you agree? Explain.

(8) Students who did a summer internship earn \$8,000 more in their first job than students who did not. Let $D_i = 1$ if student i did an internship, and let Y_{1i} and Y_{0i} be the student's first-job salary with and without one.

(a) Write $\mathsf{E}[Y_i|D_i = 1] - \mathsf{E}[Y_i|D_i = 0]$ as the effect of the internship on those who did one plus selection bias.

(b) Is the selection bias likely positive or negative? Does \$8,000 overstate or understate the effect of an internship? Explain in words.

(9) Using a random sample of 1,000 formal jobs in Brazil, we regress the log of monthly pay on years of schooling, `educ`, and get

$$\widehat{\ln(\text{pay})} = 7.21 + 0.063 \times \text{educ}.$$

(a) Interpret the slope.

(b) By about how much does predicted pay differ between a worker with a high school degree (12 years) and one with a college degree (16 years)?

(c) Give one reason to use ln(pay) rather than pay in reais.

(10) With data for 2024, a researcher regresses monthly pay on each worker's age and year of birth, both in years. Stata drops one of the two regressors.

(a) Why?

(b) The researcher wants "the effect of age, holding year of birth fixed." Can any data from a single year answer this question? Explain.

(11) True or false? Explain your answer in one or two sentences. Both statements are about the regression from class of a college's median earnings on its average SAT score (988 colleges, $\hat{\beta}_1 = 75.6$, standard error 2.94).

(a) Median earnings are skewed: a few colleges have very high earnings. So we cannot use the normal distribution to build a 95% confidence interval for β_1.

(b) In a randomized experiment in which everyone assigned to the treatment receives it, the difference in mean outcomes between the treatment and control groups is an unbiased estimator of the average causal effect of the treatment.

2. Long Question

We use data on 7,098 full-time workers aged 25 to 34 from the 2015 Current Population Survey (the data from Problem Set 1). `lahe` is the natural log of average hourly earnings, `female` equals 1 for women, and `bachelor` equals 1 for workers with a bachelor's degree and 0 for workers with a high school diploma. Panels A to D of Figure 2 show the Stata output for each regression, followed by the same regression in R.

(12) Panel A: interpret the coefficient on `female` and the constant.

(13) Panel A: test the hypothesis that there is no gender gap in pay at the 5% level. Then test the hypothesis that women earn 10% less than men.

(14) Panel B: interpret the coefficient on `female`. Before you look at Panel C, predict whether adding `bachelor` to the regression in Panel A will make the coefficient on `female` larger or smaller in absolute value. Explain.

(15) Panel C: interpret the coefficients on `female` and on `bachelor`. Then use Panels B and C to calculate the omitted variable bias in the coefficient on `female` in Panel A, and check that it explains the difference between Panels A and C.

(16) Panel D: what is the gender gap in pay among workers without a bachelor's degree? Among workers with one? Is the difference between these two gaps statistically significant at the 5% level?

(17) A journalist reads Panel C and writes: "Women in the US are paid 18% less than men for the same work." Name one omitted variable that could still bias the coefficient on `female`, and say which way.

Figure 1. STATA and R outputs

Panel A: Regression for questions (12) and (13)

```
. regress lahe female, robust
Linear regression                               Number of obs     =      7,098
                                                F(1, 7096)        =      78.75
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0107
                                                Root MSE          =     .53365
------------------------------------------------------------------------------
             |               Robust
        lahe | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |   -.112629   .0126917    -8.87   0.000    -.1375085   -.0877496
       _cons |   2.959775   .0085413   346.53   0.000     2.943031    2.976518
------------------------------------------------------------------------------
model_a <- lm(lahe ~ female, data = cps2015)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
t test of coefficients:
             Estimate Std. Error  t value  Pr(>|t|)
(Intercept)  2.9597749  0.0085413 346.5263 < 2.2e-16 ***
female      -0.1126290  0.0126917  -8.8742 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Panel B: Regression for question (14)

```
. regress bachelor female, robust
Linear regression                               Number of obs     =      7,098
                                                F(1, 7096)        =     161.66
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0221
                                                Root MSE          =     .49385
------------------------------------------------------------------------------
             |               Robust
    bachelor | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |   .1505655   .0118419    12.71   0.000     .1273519    .1737791
       _cons |   .4631554   .0077518    59.75   0.000     .4479596    .4783511
------------------------------------------------------------------------------
model_b <- lm(bachelor ~ female, data = cps2015)
coeftest(model_b, vcov = vcovHC(model_b, type = "HC1"))
t test of coefficients:
            Estimate Std. Error t value  Pr(>|t|)
(Intercept) 0.4631554  0.0077518  59.748 < 2.2e-16 ***
female      0.1505655  0.0118419  12.715 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Panel C: Regression for question (15)

```
. regress lahe female bachelor, robust
Linear regression                               Number of obs     =      7,098
                                                F(2, 7095)        =     838.23
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1916
                                                Root MSE          =     .48245
------------------------------------------------------------------------------
             |               Robust
        lahe | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |  -.1821928   .0116014   -15.70   0.000     -.204935   -.1594506
    bachelor |   .4620164   .0115786    39.90   0.000     .4393189    .4847139
       _cons |    2.74579   .0090747   302.58   0.000        2.728    2.763579
------------------------------------------------------------------------------
model_c <- lm(lahe ~ female + bachelor, data = cps2015)
coeftest(model_c, vcov = vcovHC(model_c, type = "HC1"))
t test of coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept)  2.7457895  0.0090747 302.576 < 2.2e-16 ***
female      -0.1821928  0.0116014 -15.704 < 2.2e-16 ***
bachelor     0.4620165  0.0115786  39.903 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Panel D: Regression for question (16)

```
. regress lahe female bachelor female_bach, robust
Linear regression                               Number of obs     =      7,098
                                                F(3, 7094)        =     571.37
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1916
                                                Root MSE          =     .48247
------------------------------------------------------------------------------
             |               Robust
        lahe | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |  -.1893346   .0162496   -11.65   0.000    -.2211887   -.1574805
    bachelor |   .4567787   .0157076    29.08   0.000     .4259872    .4875702
 female_bach |   .0129219   .0230596     0.56   0.575    -.0322817    .0581256
       _cons |   2.748215   .0100929   272.29   0.000      2.72843       2.768
------------------------------------------------------------------------------
model_d <- lm(lahe ~ female + bachelor + female_bach, data = cps2015)
coeftest(model_d, vcov = vcovHC(model_d, type = "HC1"))
t test of coefficients:
             Estimate Std. Error  t value Pr(>|t|)
(Intercept)  2.748215   0.010093 272.2926   <2e-16 ***
female      -0.189335   0.016250 -11.6516   <2e-16 ***
bachelor     0.456779   0.015708  29.0802   <2e-16 ***
female_bach  0.012922   0.023060   0.5604   0.5752
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```